reusable-extensible-python-functions-financial-data-analysis

September 15, 2026

1 Reusable And Extensible Python Functions For Financial Data Analysis

1.1 Python Imports

[1]:
```
# Standard Library
import os
import sys
import warnings

from pathlib import Path

# Suppress warnings
warnings.filterwarnings("ignore")
```

[2]:
```
# Add the source subdirectory to the system path to allow import config from␣
 ↪settings.py
current_directory = Path(os.getcwd())
website_base_directory = current_directory.parent.parent.parent
src_directory = website_base_directory / "src"
sys.path.append(str(src_directory)) if str(src_directory) not in sys.path else␣
 ↪None

# Import settings.py
from settings import config

# Add configured directories from config to path
SOURCE_DIR = config("SOURCE_DIR")
sys.path.append(str(Path(SOURCE_DIR))) if str(Path(SOURCE_DIR)) not in sys.path␣
 ↪else None

# Add other configured directories
BASE_DIR = config("BASE_DIR")
CONTENT_DIR = config("CONTENT_DIR")
POSTS_DIR = config("POSTS_DIR")
PAGES_DIR = config("PAGES_DIR")
PUBLIC_DIR = config("PUBLIC_DIR")
```

```
SOURCE_DIR = config("SOURCE_DIR")
DATA_DIR = config("DATA_DIR")
DATA_MANUAL_DIR = config("DATA_MANUAL_DIR")

# Print system path
for i, path in enumerate(sys.path):
    print(f"{i}: {path}")
```

```
0: /usr/lib/python313.zip
1: /usr/lib/python3.13
2: /usr/lib/python3.13/lib-dynload
3:
4: /home/jared/python-virtual-envs/general-venv-p313/lib/python3.13/site-
packages
5: /home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/src
```

1.2 Track Index Dependencies

[3]:
```
# Create file to track markdown dependencies
dep_file = Path("index_dep.txt")
dep_file.write_text("")
```

[3]:
```
0
```

1.3 Python Functions

[4]:
```
from export_track_md_deps import export_track_md_deps
```

[5]:
```
from backtest_target_weight_portfolio import backtest_target_weight_portfolio
code = Path(SOURCE_DIR / "backtest_target_weight_portfolio.py").read_text()
# Copy this <!-- INSERT_backtest_target_weight_portfolio_HERE --> to index_temp.
 ↪md
export_track_md_deps(dep_file=dep_file,␣
 ↪md_filename="backtest_target_weight_portfolio.md", content=code,␣
 ↪output_type="python")
```

```
Exported and tracked: backtest_target_weight_portfolio.md
```

[6]:
```
from bb_clean_data import bb_clean_data
code = Path(SOURCE_DIR / "bb_clean_data.py").read_text()
# Copy this <!-- INSERT_bb_clean_data_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="bb_clean_data.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: bb_clean_data.md
```

[7]:
```
from build_index import build_index
code = Path(SOURCE_DIR / "build_index.py").read_text()
```

```
# Copy this <!-- INSERT_build_index_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="build_index.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: build_index.md
```

[8]:
```
from calc_fed_cycle_asset_performance import calc_fed_cycle_asset_performance
code = Path(SOURCE_DIR / "calc_fed_cycle_asset_performance.py").read_text()
# Copy this <!-- INSERT_calc_fed_cycle_asset_performance_HERE --> to index_temp.
 ↪md
export_track_md_deps(dep_file=dep_file,␣
 ↪md_filename="calc_fed_cycle_asset_performance.md", content=code,␣
 ↪output_type="python")
```

```
Exported and tracked: calc_fed_cycle_asset_performance.md
```

[9]:
```
from calc_vix_trade_pnl import calc_vix_trade_pnl
code = Path(SOURCE_DIR / "calc_vix_trade_pnl.py").read_text()
# Copy this <!-- INSERT_calc_vix_trade_pnl_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="calc_vix_trade_pnl.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: calc_vix_trade_pnl.md
```

[10]:
```
from coinbase_fetch_available_products import coinbase_fetch_available_products
code = Path(SOURCE_DIR / "coinbase_fetch_available_products.py").read_text()
# Copy this <!-- INSERT_coinbase_fetch_available_products_HERE --> to␣
 ↪index_temp.md
export_track_md_deps(dep_file=dep_file,␣
 ↪md_filename="coinbase_fetch_available_products.md", content=code,␣
 ↪output_type="python")
```

```
Exported and tracked: coinbase_fetch_available_products.md
```

[11]:
```
from coinbase_fetch_full_history import coinbase_fetch_full_history
code = Path(SOURCE_DIR / "coinbase_fetch_full_history.py").read_text()
# Copy this <!-- INSERT_coinbase_fetch_full_history_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file,␣
 ↪md_filename="coinbase_fetch_full_history.md", content=code,␣
 ↪output_type="python")
```

```
Exported and tracked: coinbase_fetch_full_history.md
```

[12]:
```
from coinbase_fetch_historical_candles import coinbase_fetch_historical_candles
code = Path(SOURCE_DIR / "coinbase_fetch_historical_candles.py").read_text()
# Copy this <!-- INSERT_coinbase_fetch_historical_candles_HERE --> to␣
 ↪index_temp.md
```

```
export_track_md_deps(dep_file=dep_file,␣
 ↪md_filename="coinbase_fetch_historical_candles.md", content=code,␣
 ↪output_type="python")
```

```
Exported and tracked: coinbase_fetch_historical_candles.md
```

[13]:
```
from coinbase_pull_data import coinbase_pull_data
code = Path(SOURCE_DIR / "coinbase_pull_data.py").read_text()
# Copy this <!-- INSERT_coinbase_pull_data_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="coinbase_pull_data.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: coinbase_pull_data.md
```

[14]:
```
from create_ma_trend_signals import create_ma_trend_signals
code = Path(SOURCE_DIR / "create_ma_trend_signals.py").read_text()
# Copy this <!-- INSERT_create_ma_trend_signals_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="create_ma_trend_signals.
 ↪md", content=code, output_type="python")
```

```
Exported and tracked: create_ma_trend_signals.md
```

[15]:
```
from df_info import df_info
code = Path(SOURCE_DIR / "df_info.py").read_text()
# Copy this <!-- INSERT_df_info_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="df_info.md", content=code,␣
 ↪output_type="python")
```

```
Exported and tracked: df_info.md
```

[16]:
```
from df_info_markdown import df_info_markdown
code = Path(SOURCE_DIR / "df_info_markdown.py").read_text()
# Copy this <!-- INSERT_df_info_markdown_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="df_info_markdown.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: df_info_markdown.md
```

[17]:
```
from export_track_md_deps import export_track_md_deps
code = Path(SOURCE_DIR / "export_track_md_deps.py").read_text()
# Copy this <!-- INSERT_export_track_md_deps_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="export_track_md_deps.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: export_track_md_deps.md
```

[18]:
```
from load_data import load_data
code = Path(SOURCE_DIR / "load_data.py").read_text()
# Copy this <!-- INSERT_load_data_HERE --> to index_temp.md
```

```
export_track_md_deps(dep_file=dep_file, md_filename="load_data.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: load_data.md
```

[19]:
```
from pandas_set_decimal_places import pandas_set_decimal_places
code = Path(SOURCE_DIR / "pandas_set_decimal_places.py").read_text()
# Copy this <!-- INSERT_pandas_set_decimal_places_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="pandas_set_decimal_places.
 ↪md", content=code, output_type="python")
```

```
Exported and tracked: pandas_set_decimal_places.md
```

[20]:
```
from plot_bar_returns_ffr_change import plot_bar_returns_ffr_change
code = Path(SOURCE_DIR / "plot_bar_returns_ffr_change.py").read_text()
# Copy this <!-- INSERT_plot_bar_returns_ffr_change_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file,␣
 ↪md_filename="plot_bar_returns_ffr_change.md", content=code,␣
 ↪output_type="python")
```

```
Exported and tracked: plot_bar_returns_ffr_change.md
```

[21]:
```
from plot_histogram import plot_histogram
code = Path(SOURCE_DIR / "plot_histogram.py").read_text()
# Copy this <!-- INSERT_plot_histogram_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="plot_histogram.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: plot_histogram.md
```

[22]:
```
from plot_scatter import plot_scatter
code = Path(SOURCE_DIR / "plot_scatter.py").read_text()
# Copy this <!-- INSERT_plot_scatter_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="plot_scatter.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: plot_scatter.md
```

[23]:
```
from plot_scatter_regression_ffr_vs_returns import␣
 ↪plot_scatter_regression_ffr_vs_returns
code = Path(SOURCE_DIR / "plot_scatter_regression_ffr_vs_returns.py").
 ↪read_text()
# Copy this <!-- INSERT_plot_scatter_regression_ffr_vs_returns_HERE --> to␣
 ↪index_temp.md
export_track_md_deps(dep_file=dep_file,␣
 ↪md_filename="plot_scatter_regression_ffr_vs_returns.md", content=code,␣
 ↪output_type="python")
```

```
Exported and tracked: plot_scatter_regression_ffr_vs_returns.md
```

[24]:
```
from plot_stats import plot_stats
code = Path(SOURCE_DIR / "plot_stats.py").read_text()
# Copy this <!-- INSERT_plot_stats_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="plot_stats.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: plot_stats.md
```

[25]:
```
from plot_time_series import plot_time_series
code = Path(SOURCE_DIR / "plot_time_series.py").read_text()
# Copy this <!-- INSERT_plot_time_series_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="plot_time_series.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: plot_time_series.md
```

[26]:
```
from plot_vix_with_trades import plot_vix_with_trades
code = Path(SOURCE_DIR / "plot_vix_with_trades.py").read_text()
# Copy this <!-- INSERT_plot_vix_with_trades_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="plot_vix_with_trades.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: plot_vix_with_trades.md
```

[27]:
```
from polygon_fetch_full_history import polygon_fetch_full_history
code = Path(SOURCE_DIR / "polygon_fetch_full_history.py").read_text()
# Copy this <!-- INSERT_polygon_fetch_full_history_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="polygon_fetch_full_history.
 ↪md", content=code, output_type="python")
```

```
Exported and tracked: polygon_fetch_full_history.md
```

[28]:
```
from polygon_fetch_ticker_details import polygon_fetch_ticker_details
code = Path(SOURCE_DIR / "polygon_fetch_ticker_details.py").read_text()
# Copy this <!-- INSERT_polygon_fetch_ticker_details_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file,␣
 ↪md_filename="polygon_fetch_ticker_details.md", content=code,␣
 ↪output_type="python")
```

```
Exported and tracked: polygon_fetch_ticker_details.md
```

[29]:
```
from polygon_pull_data import polygon_pull_data
code = Path(SOURCE_DIR / "polygon_pull_data.py").read_text()
# Copy this <!-- INSERT_polygon_pull_data_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="polygon_pull_data.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: polygon_pull_data.md
```

[30]:
```
from run_regression import run_regression
code = Path(SOURCE_DIR / "run_regression.py").read_text()
# Copy this <!-- INSERT_run_regression_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="run_regression.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: run_regression.md
```

[31]:
```
from strategy_harry_brown_perm_port import strategy_harry_brown_perm_port
code = Path(SOURCE_DIR / "strategy_harry_brown_perm_port.py").read_text()
# Copy this <!-- INSERT_strategy_harry_brown_perm_port_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file,␣
 ↪md_filename="strategy_harry_brown_perm_port.md", content=code,␣
 ↪output_type="python")
```

```
Exported and tracked: strategy_harry_brown_perm_port.md
```

[32]:
```
from summary_stats import summary_stats
code = Path(SOURCE_DIR / "summary_stats.py").read_text()
# Copy this <!-- INSERT_summary_stats_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="summary_stats.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: summary_stats.md
```

[33]:
```
from yf_pull_data import yf_pull_data
code = Path(SOURCE_DIR / "yf_pull_data.py").read_text()
# Copy this <!-- INSERT_yf_pull_data_HERE --> to index_temp.md
export_track_md_deps(dep_file=dep_file, md_filename="yf_pull_data.md",␣
 ↪content=code, output_type="python")
```

```
Exported and tracked: yf_pull_data.md
```